6.2.2 Series and parallel circuits

Content	Key opportunities for skills development
There are two ways of joining electrical components, in series and in parallel. Some circuits include both series and parallel parts. For components connected in series: • there is the same current through each component • the total potential difference of the power supply is shared between the components • the total resistance of two components is the sum of the resistance of each component.	

Content	Key opportunities for skills development
$R_{total} = R_1 + R_2$ resistance, R, in ohms, Ω For components connected in parallel: • the potential difference across each component is the same • the total current through the whole circuit is the sum of the currents through the separate components • the total resistance of two resistors is less than the resistance of the smallest individual resistor. Students should be able to:	MS 1c, 3b, 3c, 3d
• use circuit diagrams to construct and check series and parallel circuits that include a variety of common circuit components • describe the difference between series and parallel circuits • explain qualitatively why adding resistors in series increases the total resistance whilst adding resistors in parallel decreases the total resistance	AT 7
• explain the design and use of dc series circuits for measurement and testing purposes	WS 1.4
• calculate the currents, potential differences and resistances in dc series circuits • solve problems for circuits which include resistors in series using the concept of equivalent resistance. Students are **not** required to calculate the total resistance of two resistors joined in parallel.	MS 1c, 3b, c, d